

Cover Letter: IEEE Transactions Submission

Date: September 21, 2026

To:

The Editor-in-Chief

IEEE Transactions on Systems, Man, and Cybernetics: Systems (or *IEEE Transactions on Cybernetics*)

Subject: Manuscript Submission — *Generation, Survival, and Evolution of Open Complex Giant Systems: Inference Deductions and Isomorphic Proof of OCGS Based on the Non-IID First Principle*

Dear Editor-in-Chief and Associate Editors,

I am pleased to submit our original research manuscript titled “**Generation, Survival, and Evolution of Open Complex Giant Systems: Inference Deductions and Isomorphic Proof of OCGS Based on the Non-IID First Principle**” for consideration for publication in *IEEE Transactions*.

Research Significance and Core Contributions

The long-term survival, governance, and autonomous closed-loop computability of Open Complex Giant Systems (OCGS)—first conceptualized by Academician Qian Xuesen in the 1990s—represent fundamental open questions in system science and complexity science. While Prof. Longbing Cao (*IEEE TKDE 2014, IEEE SMC 2009*) established the Non-Independent and Non-Identically Distributed (Non-IID) theoretical foundation for data science and complex learning, formal mathematical computability proofs and large-scale industrial validation for universal OCGS have remained elusive.

This paper bridges this gap by decoupling qualitative cybernetic deductions from formal quantitative proofs:

1. Non-IID First Principle & Formal Computability Guarantee:

By modeling high-dimensional non-convex state spaces as continuous topological manifolds $\mathcal{M}$, we introduce an orthogonal projection operator $\mathcal{P}_{\text{orthogonal}}$ that algebraically prunes combinatorial state complexity from $O(N!)$ to polynomial $O(N^k)$ (Lemma 1). We construct a complete Banach space $\mathcal{B}$ with normalized structural norms and establish the **Structure/Operator Fixed Point Theorem** (Theorem 1), proving that under Lipschitz contraction ($\gamma < 1$), the governance algorithm structure automatically converges to a unique, self-consistent fixed point $\mathcal{S}^*$.

2. 1-to-1 Isomorphism to Qian Xuesen’s OCGS Theory:

We construct a rigorous 1-to-1 physical and cybernetic isomorphic mapping between eight qualitative deductions (covering Teleology, Essence, Methodology, Capability, Mechanism, Pathway, Dynamics, and Evolution) and Qian’s metasyntesis methodology (HWME Hall of Workshop, Department of Overall Design, Metasynthetic Wisdom).

3. 18-Year Industrial Validation & “Human-Out-of-the-Loop” Autonomy:

The theoretical framework is grounded in 18 years of continuous industrial physical practice (Lenovo Group IPS System) and the IPC physics engine (20 patents filed, 5 granted). Operating on global supply networks controlling 50,000 discrete demands, 2,000,000 SKU-Site nodes, and 150,000 topological constraints, the system maintains an autonomous decision rate of **over 95% human-out-of-the-loop** with 48-hour response rates $> 98\%$.

4. Hardcore Popperian Falsifiability & Cross-Domain Metrics:

We establish strict Popperian falsifiability boundaries for all eight deductions, complete with an alignment matrix of observable metric indicators across four major complex giant systems: **Meteorology, Urban Traffic, Macroeconomics, and Enterprise Supply Networks.**

IEEE Double-Blind Review Compliance

In full accordance with IEEE double-blind peer review guidelines: - The submitted manuscript (Paper_2_IEEE_Double_Blind_EN) has been thoroughly anonymized, removing author names, institutional affiliations, contact emails, ORCID identifiers, and self-identifying repository links. - All complete metadata, author biographies, institutional affiliations, and repository URLs are provided in the separate **Title Page** (Paper_2_IEEE_Title_Page). - This manuscript represents original work that has not been published previously and is not under consideration for publication elsewhere.

Thank you very much for your time, consideration, and leadership in processing our submission. We look forward to receiving the reviewers' constructive feedback.

Sincerely,

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